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Epistemic Constraints and Semantic Compression in Natural Language Processing: A Theoretical Foundation for the HGC³AE² Framework (v1.1 errata)

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2026

Contents

1. Introduction

The governance-first architecture proposed in Paper One rests on a structural argument: that agentic systems fail not because they lack intelligence, but because they lack sufficient alignment across the governance, context, collaboration, and execution layers that bind human intent to machine behavior.[1] That argument is sound as far as it goes. What it does not supply — and what the present paper addresses — is an account of the epistemic substrate from which those failures emerge. **Confident misalignment**, Paper One's central failure condition, is a behavioral observation: the system produces output that is coherent, plausible, and authoritative, yet wrong relative to verified state or user intent. The behavioral account is necessary but not sufficient. A complete theory of confident misalignment requires an account of why statistical inference, operating at scale on large language models, reliably produces this condition when it crosses certain boundaries. That account is the contribution of this paper.

The existing literature on AI safety and NLP failure modes is rich in taxonomies of what goes wrong. Structural failure modes — hallucination, context drift, reward hacking, proxy substitution — have been documented with increasing precision. Standards bodies have developed governance frameworks that catalog risk surfaces and escalation requirements in substantial detail.[2] What is largely absent from this literature is a mechanistic account of why language systems fail at domain boundaries in the specific way they do: not randomly, not corruptly, but with confident fluency and systematic misdirection. The answer lies neither in the architecture per se nor in the volume of training data. It lies in the epistemic structure of language itself, and in the relationship between how language systems compress meaning and how meaning is constituted in the domains those systems are being asked to serve. The failure is not incidental to the technology; it is structurally generated by it.

Consider a query about whether a specific event qualifies as *force majeure* under the governing clause of a commercial contract. The query is formally well-formed, the language is unambiguous, and a large language model has access to substantial distributional data about force majeure doctrine across jurisdictions and contract types. Yet the answer to this query is not determined by which response is statistically most likely given the training distribution. It is determined by jurisdiction, by the specific clause language, by applicable case law, and ultimately by an act of adjudicative selection within a legal framework whose authority derives from institutional processes that no distributional model encodes. The fluent, confident response that a language system produces in this case may satisfy every distributional measure of plausibility and fail every legal standard of validity. That disjunction is not a retrieval problem. It is an epistemic one.

Consider, in parallel, a clinical risk-stratification system that assigns patients to high- and low-risk categories to guide treatment prioritization. The system has been trained on population health data and produces outputs that correlate with aggregate outcomes. What it cannot supply — without explicit architectural intervention — is the clinical judgment that a trained physician exercises when evaluating an algorithmic output against a specific patient's presenting condition, comorbidities, and treatment history. The algorithm operates within a constraint structure defined by statistical association; the physician operates within a constraint structure defined by clinical protocol, professional training, and an individual patient relationship. When those two constraint structures are conflated — when the statistical output is treated as equivalent to a clinically valid judgment — the result is not error in the ordinary sense. It is a substitution of one epistemic framework for another, with consequences the empirical record has documented in detail.

This paper advances three related claims. First, it introduces the concept of **thin context**

[1] Justin H. Kuiper, CISSP, "Mitigating Confident Misalignment in Agentic Systems: The HGC³AE² Framework," Non Sequitur Publishing, 2026, v1.0-preprint, SHA 6e0b127f, https://nonsequitur.tech/white-papers/hgc3ae2/.

[2] National Institute of Standards and Technology, *Artificial Intelligence Risk Management Framework (AI RMF 1.0)*, NIST AI 100-1 (Gaithersburg, MD, 2023), https://doi.org/10.6028/NIST.AI.100-1.

as a formally defined epistemic condition: the state in which a representation or prompt lacks the domain-specific constraints necessary for a validation mechanism to constitute an authoritative answer. Thin context is structurally distinct from ambiguity, underspecification, and epistemic uncertainty, and it receives full formal treatment in §4. Second, it identifies **semantic compression** as the generative mechanism of thin context: the process by which domain-specific epistemic constraints are systematically collapsed during language representation, such that the resulting encoding carries distributional statistical properties but not the validation-mechanism-specific information required for authoritative inference. That mechanism is the subject of §5. Third, the paper argues in §7 that thin context is the epistemic precondition beneath every structural failure mode that Paper One identifies — and that this causal relationship establishes the epistemic necessity, not merely the governance prudence, of the HGC³AE² framework's architecture.

The evidence from human-AI interaction research reinforces this account from a different direction. The guidelines developed for human-AI interaction document the conditions under which collaboration succeeds and fails, and the failure conditions cluster around the system's behavior under changing task contexts, under conditions of domain-specific uncertainty, and when user expertise exceeds the system's capacity for domain-sensitive inference.[3] Trust calibration research shows that human reliance on automated systems is mediated by domain-specific expectations of competence, and that miscalibrated trust is most dangerous precisely when systems perform with high apparent confidence in conditions where their outputs are least reliable.[4] These findings describe the behavioral surface of a problem whose structure this paper locates at the level of epistemic architecture. Human disambiguation succeeds where statistical inference fails for reasons that are not incidental — they are structural, and §6 examines them in full.

The present paper does not re-derive or re-describe the HGC³AE² framework. That work belongs to Paper One. What this paper provides is the epistemic foundation that makes the framework's structural requirements legible as necessities rather than precautions. When the architecture of language representation is understood as a systematic compressor of domain-specific epistemic constraints, the governance-first insistence on human authority, curated context, and formal reconciliation processes follows not from a preference for caution but from an analysis of what authoritative inference actually requires. The argument begins with the mechanics of compression.

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2. Language as Compression: Information-Theoretic Foundations

Every system for representing the meaning of language is, at a fundamental level, a compression system. The compressor takes the indefinitely rich domain of meaning — with all its contextual dependencies, validation requirements, and domain-specific constraints — and produces a representation that is more compact, more portable, and more computationally tractable. What the compressor discards in doing so is not random. Compression selects for statistical regularity and generalizability; it systematically devalues what is rare, domain-specific, and constraint-dense. This is not a flaw in any particular representation scheme.

[3] Saleema Amershi, Dan Weld, Mihaela Vorvoreanu, Adam Fourney, Besmira Nushi, Penny Collisson, Jina Suh, Shamsi Iqbal, Paul N. Bennett, Kori Inkpen, Jaime Teevan, Ruth Kikin-Gil, and Eric Horvitz, "Guidelines for Human-AI Interaction," in *Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems* (2019), https://doi.org/10.1145/3290605.3300233.

[4] Kevin Anthony Hoff and Masooda Bashir, "Trust in Automation: Integrating Empirical Evidence on Factors That Influence Trust," *Human Factors* 57, no. 3 (2015): 407–434, https://doi.org/10.1177/0018720814547570.

It is the defining property of compression as such, recognized in its most rigorous form by Shannon seven decades before large language models made its consequences unavoidable.

Shannon's 1948 formalization of information as a measurable quantity established that any communication channel has a finite capacity, and that transmitting information through a channel of bounded capacity necessarily involves decisions about what to encode precisely and what to discard.[5] The noiseless coding theorem demonstrates that optimal codes compress information toward the entropy of the source distribution — which means that events with low probability in the training distribution are encoded inefficiently or not at all. The implications for language representation are direct: a model trained on a corpus in which legal adjudicative reasoning appears rarely and general conversational English appears frequently will encode the former with far less representational fidelity than the latter. The compression is not malicious; it is mathematically optimal given the objective. The objective, however, is statistical generalizability, not domain constraint preservation, and the two are in structural tension.

The distributional hypothesis, formalized by Harris in 1954, gave that statistical generalizability a theoretical grounding that subsequent decades of computational linguistics converted into a research program.[6] Harris's central claim — that linguistic items with similar distributional environments have similar meanings — provided the epistemological foundation for treating meaning as extractable from co-occurrence patterns. This was a productive simplification. It is also, at its foundations, a stipulation that the meaning of a term is whatever can be inferred from its statistical neighborhood in the corpus. What this conception captures is substantial: relative synonymy, semantic field membership, pragmatic register, tonal and stylistic similarity. What it systematically omits is the relationship between linguistic form and the validation mechanisms through which meaning is constituted in specific domains of practice. A term's distributional neighborhood in a general corpus provides no information about whether a given use of that term satisfies the evidentiary requirements of a legal argument, the clinical protocol governing a treatment decision, or the tolerance specification constraining an engineering judgment.

Bengio and colleagues' neural probabilistic language model, introduced in 2003, operationalized the distributional hypothesis as a learned function mapping word sequences to probability distributions over continuations, with meaning encoded as position in a shared low-dimensional space.[7] The critical move was the discovery that this continuous representational space enabled generalization across surface-distinct but semantically related contexts — a genuine advance over discrete n-gram methods. It was also the founding instantiation of a compression regime: words were represented by dense vectors whose dimensions encoded statistically useful distinctions and whose information capacity was fixed in advance. The domain-specific constraint structure of any given word's use could not survive this compression unless it correlated with high-frequency patterns in the training distribution.

The static vector embedding architectures that followed — Mikolov and colleagues' distributed representations of words and phrases, Pennington, Socher, and Manning's global vector model — realized the distributional hypothesis at scale with explicit geometric interpretability.[8] [9] Words became points in a high-dimensional space; semantic proximity became

[5] Claude E. Shannon, "A Mathematical Theory of Communication," *Bell System Technical Journal* 27, no. 3 (July 1948): 379–423, https://doi.org/10.1002/j.1538-7305.1948.tb01338.x.

[6] Zellig S. Harris, "Distributional Structure," *Word* 10, no. 2–3 (1954): 146–162, https://doi.org/10.1080/00437956.1954.11659520.

[7] Yoshua Bengio, Réjean Ducharme, Pascal Vincent, and Christian Jauvin, "A Neural Probabilistic Language Model," *Journal of Machine Learning Research* 3 (2003): 1137–1155.

[8] Tomas Mikolov, Ilya Sutskever, Kai Chen, Greg Corrado, and Jeffrey Dean, "Distributed Representations of Words and Phrases and their Compositionality," in *Advances in Neural Information Processing Systems 26* (NIPS 2013), 3111–3119, https://papers.nips.cc/paper/2013/hash/9aa42b31882ec039965f3c4923ce901b-Abstract.html.

[9] Jeffrey Pennington, Richard Socher, and Christopher D. Manning, "GloVe: Global Vectors for Word Representation," in *Proceedings of the 2014 Conference on Empirical Methods in Natural Language Processing (EMNLP)*,

spatial proximity; and algebraic operations in that space — vector addition, cosine similarity — became proxy operations on meaning. The representational gains were substantial and well-documented across benchmark tasks. So were the losses. Polysemy collapsed: a single vector represented all senses of a word, weighted by their frequency in the training distribution. The specific sense of *contract* operative in a *force majeure* clause and the general conversational sense of *contract* as agreement compressed to a single point, positioned wherever the aggregate distributional evidence placed it. The compression is geometrically elegant. The epistemic loss is structural: once a term's domain-specific sense has been compressed into a single weighted vector, the domain-specific constraints governing that sense's valid application are no longer recoverable from the representation.

Sequential architectures introduced temporal context into language representation, with long short-term memory networks providing a mechanism for maintaining state across longer dependencies than earlier recurrent models could sustain.[10] This was a meaningful advance in capturing syntactic structure, discourse-level coherence, and coreference across paragraphs. It did not address the domain constraint problem, because domain-specific constraints are not primarily a function of sequential position in a document. They are properties of the institutional and epistemic context in which language is being used — the validation framework within which a claim is being made and assessed — and that context is not recoverable from sequential window expansion at any scale.

The transformer architecture and the contextualized representations it enables — instantiated most influentially in BERT's bidirectional pre-training regime — introduced self-attention as a learned compression function over the full context window, producing representations that vary with the surrounding text rather than collapsing to a single position in a fixed embedding space.[11] [12] Where static embeddings gave one vector per word type regardless of usage context, attention produces representations that shift with the sentential environment. This is a genuine improvement in the granularity of semantic compression. What attention attends to, however, is distributional co-occurrence — which tokens appear with which other tokens, and in what syntactic and semantic patterns, across the training distribution. Attention does not and cannot attend to domain validation requirements, because those requirements are not encoded in distributional statistics. The contextualized representation of *force majeure* in a document that resembles a commercial contract will reflect patterns from other contract-like documents. It will not reflect the adjudicative framework that determines whether a specific factual scenario satisfies the clause's legal conditions under governing law.

Scale has not resolved this tension; it has extended it in a specific and important direction. Larger language models trained on larger and more diverse corpora generalize more broadly, perform better across a wider range of tasks, and produce more fluent output in more domains.[13] The distributional representation of any given domain improves as the training distribution more fully samples that domain's surface patterns. What does not improve with scale

1532–1543 (Doha, Qatar: Association for Computational Linguistics, 2014), https://doi.org/10.3115/v1/D14-1162.

[10] Sepp Hochreiter and Jürgen Schmidhuber, "Long Short-Term Memory," *Neural Computation* 9, no. 8 (1997): 1735–1780, https://doi.org/10.1162/neco.1997.9.8.1735.

[11] Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N. Gomez, Łukasz Kaiser, and Illia Polosukhin, "Attention Is All You Need," in *Advances in Neural Information Processing Systems 30* (NeurIPS 2017), 5998–6008, https://papers.nips.cc/paper_files/paper/2017/hash/3f5ee243547dee91fbd053c1c4a845aa-Abstract.html.

[12] Jacob Devlin, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova, "BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding," in *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1*, 4171–4186 (Minneapolis: Association for Computational Linguistics, 2019), https://doi.org/10.18653/v1/N19-1423.

[13] Tom B. Brown, Benjamin Mann, Nick Ryder, Melanie Subbiah, Jared Kaplan, Prafulla Dhariwal, Arvind Neelakantan, Pranav Shyam, Girish Sastry, Amanda Askell, Sandhini Agarwal, Ariel Herbert-Voss, Gretchen Krueger, Tom Henighan, Rewon Child, Aditya Ramesh, Daniel M. Ziegler, Jeffrey Wu, Clemens Winter, Christopher Hesse, Mark Chen, Eric Sigler, Mateusz Litwin, Scott Gray, Benjamin Chess, Jack Clark, Christopher Berner, Sam McCandlish, Alec Radford, Ilya Sutskever, and Dario Amodei, "Language Models Are Few-Shot Learners," in *Advances in Neural Information Processing Systems 33* (NeurIPS 2020), 1877–1901, https://arxiv.org/abs/2005.14165.

is the reconstruction of epistemic constraints that are structurally absent from any distributional representation — not because data volume is insufficient but because the constraints are properties of institutional and validation-mechanism contexts that co-occurrence statistics do not encode. The foundation model paradigm, wherein a single large pre-trained model is adapted to a wide range of downstream deployments, intensifies this concern: the domain constraint omissions of the foundation are inherited by every downstream application, and the high fluency of large-scale outputs makes those omissions systematically harder to detect.[14]

The appropriate analogy is Shannon's noisy channel, not the noiseless channel. In the noiseless case, optimal coding recovers the source exactly. In the noisy channel, there is an irreducible error floor determined by the channel's noise characteristics, and below a certain signal rate the receiver cannot reconstruct what the sender transmitted regardless of coding sophistication. Domain-specific epistemic constraints occupy the position of the low-rate signal in a high-noise channel: they are present in the source, systematically degraded in transmission, and not reconstructible from the received signal alone. This is not a problem with any particular model architecture, training procedure, or parameter count. It is a property of the compression regime, and compression is not incidental to language representation. It is constitutive of it. The domain-specific failures examined in §§3–5 are not engineering artifacts to be corrected with better architectures. They are structural outputs of the technology — and understanding what kind of structural outputs they are requires a prior account of what domain-specific epistemic constraints actually are.

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3. Epistemic Domains and Validation Mechanisms

The compression problem described in §2 would be manageable if language operated within a single, unified epistemic framework — if every domain constituted truth by the same procedure and the same constraints applied universally. It does not. Different fields of practice generate claims about the world by fundamentally different means, and those means are not variations on a common theme. They are structurally distinct systems for constituting what counts as true, valid, or authoritative. The concept required to make this claim precise is that of an **epistemic domain**: a field of practice in which truth is constituted by a specific, institutionally embedded procedure for evaluating claims. Law, medicine, engineering, public administration, and scientific inquiry do not merely differ in vocabulary or subject matter. They differ in the rules by which claims are made authoritative — and those rules are what distributional compression systematically cannot preserve.

Each epistemic domain is organized around what this paper terms a **validation mechanism**: the domain-specific procedure that constitutes a claim as true, false, or indeterminate within that domain's framework of practice. In law, the validation mechanism for disputed claims is adjudication — the application of statutory text, precedent, and doctrinal framework to a specific factual scenario by an authority empowered to make determinations binding within the jurisdiction. In medicine, it is clinical protocol combined with individual patient judgment: the evaluation of evidence against established treatment guidelines, modified by the specific conditions, history, and preferences of the patient under care. In engineering, it is verification against specification: the demonstration that a design or construction satisfies the tolerance and performance requirements that define success within the applicable standard. In science, it is empirical replication: the independent reproduction of an experimental result under controlled conditions by parties with no stake in the outcome. These mechanisms are not interchangeable. A claim that passes adjudicative validation is not thereby empirically

[14] Rishi Bommasani et al., "On the Opportunities and Risks of Foundation Models," arXiv:2108.07258, 2021, Stanford Center for Research on Foundation Models, https://arxiv.org/abs/2108.07258.

replicated. A claim that has been clinically validated for one patient population is not thereby validated for another.

This account has philosophical precedents of some depth. Kuhn's analysis of scientific knowledge showed that what counts as evidence, what counts as a solution, and what counts as a problem are not universally determined but paradigm-relative — constituted by the shared commitments and exemplars of a research community, and subject to crisis when those commitments conflict with accumulating anomalies.[15] The insight relevant to the present argument is not Kuhn's account of scientific revolutions specifically but the more general claim it implies: that criteria of validity are internal to epistemic frameworks, not external to them. When two frameworks with incommensurable criteria encounter the same empirical situation, the conflict cannot be resolved by appeal to some neutral standard — because the standards are themselves what differ. Translating between frameworks requires more than a vocabulary mapping. It requires a reconfiguration of what constitutes legitimate evidence and acceptable inference.

Wittgenstein's concept of language games provides a complementary analysis at the level of meaning rather than validity.[16] On this account, the meaning of a linguistic expression is constituted by its use within a form of life — the practical, institutional, and social context in which the expression is deployed and the activity in which it plays a role. A term like *negligence* does not carry a context-free meaning that legal usage merely applies; its meaning is its role in the legal language game, with all the procedural and evidentiary constraints that game imposes. The same term appearing in ordinary conversational English plays a different role, governed by different constraints, constituting a different claim. Cross-domain language use is not a translation problem with a solution; it is a case of language operating within a game whose rules are no longer being observed, with no mechanism available to the language system to detect the rule change.

Grice's analysis of conversational implicature makes the dependency on shared contextual assumptions explicit.[17] The cooperative principle, and the maxims that instantiate it, generate meaning not from the literal content of what is said but from the interaction between that content and the assumptions that speaker and hearer share about the communicative situation. Those assumptions include — crucially — assumptions about what domain of practice the exchange inhabits, what validation standards are operative, and what counts as a relevant and informative contribution. A sentence that implicates one thing in a legal deposition implicates something quite different in a clinical consultation, not because the words are different but because the domain-specific assumptions that generate implicature are different. Distributional models trained on a mixture of text from multiple domains have no access to those assumptions as such; they have only the statistical patterns of word co-occurrence across a corpus that does not distinguish between the assumptions governing legal language and those governing clinical language.

The *force majeure* example introduced in §1 can now be given its full epistemic description. A *force majeure* clause in a commercial contract — for example, a provision excluding from breach liability events "beyond the reasonable control of the party" — is not interpreted by consulting a distributional model of how similar language has been used in similar documents. It is interpreted by applying the legal validation mechanism to the specific facts: examining the governing law's definition of *force majeure*, evaluating whether the event satisfies the doctrinal tests established by applicable precedent, and making an adjudicative determination that is authoritative because it has been made by an entity empowered to make such determinations.[18] The Restatement (Second) of Contracts addresses supervening impracti-

[15] Thomas S. Kuhn, *The Structure of Scientific Revolutions*, 2nd ed. (Chicago: University of Chicago Press, 1970).

[16] Ludwig Wittgenstein, *Philosophical Investigations*, trans. G.E.M. Anscombe (Oxford: Blackwell, 1953).

[17] H. Paul Grice, "Logic and Conversation," in Peter Cole and Jerry L. Morgan, eds., *Syntax and Semantics*, vol. 3: *Speech Acts* (New York: Academic Press, 1975), 41–58.

[18] American Law Institute, *Restatement (Second) of Contracts* (Philadelphia: American Law Institute, 1981), §261

cability — the closest doctrinal concept — in terms of specific conditions (occurrence of a contingency, non-allocation of the risk by the contract, impracticability of performance) that are each contested legal conclusions, not statistical outputs. A language model producing an answer to this query is not applying this mechanism. It is producing a fluent continuation of a sequence that resembles legal text, with no means of access to the adjudicative authority that constitutes the answer as legally valid.

The clinical risk-stratification scenario lands its empirical weight here. Obermeyer and colleagues' analysis of a widely deployed commercial risk algorithm found that the algorithm used health-care cost as a proxy for health need — a proxy that correlated with need across the training population but did so differentially across racial groups, producing systematic underestimation of need in Black patients.[19] The failure is precisely what the epistemic domain account predicts: the algorithm operated within a statistical validation framework (correlation with training distribution outcomes), while the deployment context required a clinical validation framework (accurate assessment of individual patient need against the standard of care). The proxy's statistical validity was real. Its clinical validity was compromised at the domain boundary where the two frameworks diverged. No increase in scale, data volume, or model sophistication would have corrected this, because the failure was not in the model's distributional coverage but in the substitution of one validation mechanism for another.

Topol's analysis of the convergence of human and artificial intelligence in clinical medicine identifies the precise role that human authority must play in high-stakes medical contexts: not as a fail-safe for inaccurate predictions but as the irreplaceable locus of clinical validation — the process by which a population-level statistical output is evaluated against a patient-specific care context.[20] What Topol describes as a collaboration requirement is, in the terms of the present paper, an epistemic domain requirement: the clinical domain's validation mechanism cannot be computationally approximated, and human authority is required not as a preference but as the only available implementation of that mechanism. The same principle generalizes across epistemic domains. Legal adjudication cannot be replaced by distributional inference. Engineering specification verification cannot be approximated by statistical correlation. Scientific replication cannot be substituted by training distribution coverage. Each domain's validation mechanism is constitutively resistant to distributional compression — and it is precisely the constraints that implement those mechanisms that §4 will show are stripped in the generation of thin context.

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4. Thin Context: A Formal Definition

The analysis in §§2 and 3 has established two complementary facts. Language representation is a compression process that systematically trades domain-specific constraint information for statistical generalizability. Epistemic domains constitute truth through validation mechanisms whose operations require precisely the kind of constraint information that compression systematically removes. The intersection of these two facts produces a condition that requires a name and a precise definition, because naming it is the prerequisite for diagnosing it, measuring it, and addressing it architecturally. The condition at that intersection — the state in which the constraints required for domain-specific meaning-resolution have been stripped

(Discharge by Supervening Impracticability).

[19] Ziad Obermeyer, Brian Powers, Christine Vogeli, and Sendhil Mullainathan, "Dissecting Racial Bias in an Algorithm Used to Manage the Health of Populations," *Science* 366, no. 6464 (October 25, 2019): 447–453, https://doi.org/10.1126/science.aax2342.

[20] Eric J. Topol, "High-performance medicine: the convergence of human and artificial intelligence," *Nature Medicine* 25 (2019): 44–56, https://doi.org/10.1038/s41591-018-0300-7.

away, leaving distributional proximity as the sole basis for inference — is what I term **thin context**.

Thin context is defined formally as the epistemic condition in which a representation or prompt presents a query directed at a domain-specific claim while failing to supply the constraints that the relevant validation mechanism requires for authoritative resolution of that claim. Four conditions must jointly obtain. First, the query or prompt concerns a claim whose validity is constituted by a domain-specific validation mechanism — one that operates by adjudication, clinical protocol, engineering specification, empirical replication, or analogous procedure rather than by statistical approximation. Second, the representation does not supply — through explicit encoding, retrieval, or structured context — the constraints that the validation mechanism requires: the governing law, the applicable protocol, the relevant specification, or the constitutive framework. Third, a statistical language model can produce a fluent, confident, and structurally complete response to the query by distributional inference. Fourth, that response cannot be validated against the domain's constitutive procedure, because the information required for such validation was not present in the representation. All four conditions together constitute thin context. The third condition is what makes thin context dangerous; it is what distinguishes it from more visible failure modes.

Thin context is not ambiguity, and the distinction matters for how the condition is to be addressed. Ambiguity is a property of linguistic form: a word, phrase, or sentence that admits of multiple distinct meanings given the surface structure alone.[21] [22] The word *bank* is ambiguous between financial institution and river edge; the sentence *visiting relatives can be annoying* is syntactically ambiguous between two parse structures. Ambiguity at the level of form is a well-studied phenomenon in natural language processing, and disambiguation systems operate by exploiting contextual signals to select among the available readings. Thin context is not a property of form; it is a property of the epistemic situation. A query can be formally unambiguous — admitting of exactly one literal reading — and still produce thin context, if the constraints required for the validation mechanism to constitute an authoritative answer are absent from the representation. "Does this event qualify as *force majeure* under the governing clause?" is formally unambiguous. It is epistemically underequipped, because the adjudicative framework that constitutes the answer is not in the prompt.

Thin context is not underspecification, though the two are related. Underspecification is the condition in which relevant information is missing — the representation lacks facts that, if present, would determine the answer. Adding more information resolves underspecification. Thin context is the condition in which the constraints governing how information should be evaluated are missing — the representation lacks not facts but the framework within which facts acquire their bearing on the answer. The distinction is operationally significant. A language model can respond to underspecification by requesting or inferring additional information; it responds to thin context by substituting distributional plausibility for validation-mechanism authority, with no detectable difference in its output behavior. Bender and Koller's analysis of the form-meaning gap in large language models is the nearest prior articulation of this point: the capacity to produce fluent, contextually coherent text by predicting distributional continuations is not equivalent to access to the grounded meaning that those texts express, because grounded meaning depends on the relationship between form and world that distributional co-occurrence does not encode.[23]

Thin context is not uncertainty. Uncertainty is an epistemic state with respect to a propo-

[21] Roberto Navigli, "Word Sense Disambiguation: A Survey," *ACM Computing Surveys* 41, no. 2, Article 10 (February 2009): 1–69, https://doi.org/10.1145/1459352.1459355.

[22] Nancy Ide and Jean Véronis, "Introduction to the Special Issue on Word Sense Disambiguation: The State of the Art," *Computational Linguistics* 24, no. 1 (March 1998): 1–40 (ACL Anthology: J98-1001).

[23] Emily M. Bender and Alexander Koller, "Climbing towards NLU: On Meaning, Form, and Understanding in the Age of Data," in *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, 5185–5198 (Online: Association for Computational Linguistics, 2020), https://doi.org/10.18653/v1/2020.acl-main.463.

sition: the agent assigns a credence to the proposition that falls short of full belief. A well-calibrated model expressing uncertainty about a legal conclusion is not in thin context — it is in a legitimate epistemic state. Thin context is the condition in which the conditions for the proposition's being constituted as true or false within the relevant domain are absent. The difference is between "I am not sure whether P" and "the procedure through which P would be made true or false in this domain has not been executed and cannot be executed from the available representation." Liu and Ying Liu's analysis of uncertainty estimation for word sense disambiguation makes the relevant distinction empirically visible: models can express well-calibrated uncertainty about their distributional outputs while remaining systematically insensitive to the domain boundary at which distributional inference ceases to be the appropriate procedure for constituting an answer.[24]

Piantadosi, Tily, and Gibson's analysis of the communicative function of ambiguity illuminates thin context from a different angle.[25] Their core claim is that ambiguity is not a defect to be eliminated from natural language but a functional property: in communication contexts where shared assumptions and background knowledge are rich, the reuse of short, ambiguous forms is efficient because the disambiguation is handled by the context. Thin context is the pathological limit case of this dynamic. When the context that enables efficient disambiguation is stripped away — by compression, by domain crossing, by the absence of the validation-mechanism framework — ambiguity that was functional in the source context becomes an epistemic obstacle in the deployment context. The language is doing what it was designed to do; the epistemic scaffolding that made that design work has been removed.

The four diagnostic criteria can be applied directly to the *force majeure* example to confirm the diagnosis. The query concerns a claim whose validity is constituted by legal adjudication (condition one). The representation — a natural language prompt, absent governing law, applicable jurisdiction, and relevant precedent — does not supply the adjudicative framework (condition two). A language model can produce a fluent, structured, legally plausible response (condition three). That response cannot be validated as legally authoritative, because the adjudicative determination it would require has not been made by an entity empowered to make it (condition four). The diagnosis is thin context. The same analysis applies to the clinical risk-stratification case: the algorithm's output addresses a claim governed by clinical protocol (condition one), without supplying the patient-specific and protocol-specific constraints (condition two), producing confident stratification (condition three) that cannot be clinically validated for the individual patient at the domain boundary (condition four).

What thin context is not, finally, is a defect in the language model's training or a failure of deployment engineering. It is a structural output of the relationship between the compression regime described in §2 and the validation-mechanism requirements described in §3. §5 examines the mechanism by which that relationship produces thin context as a regular and predictable consequence — not as an exception to be patched but as the normal operating condition of distributional language systems deployed across epistemic domain boundaries.

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[24] Zhu Liu and Ying Liu, "Ambiguity Meets Uncertainty: Investigating Uncertainty Estimation for Word Sense Disambiguation," in *Findings of the Association for Computational Linguistics: ACL 2023*, 3963–3977 (Toronto, Canada: Association for Computational Linguistics, 2023), https://doi.org/10.18653/v1/2023.findings-acl.245.

[25] Steven T. Piantadosi, Harry Tily, and Edward Gibson, "The Communicative Function of Ambiguity in Language," *Cognition* 122, no. 3 (March 2012): 280–291, https://doi.org/10.1016/j.cognition.2011.10.013.

5. Semantic Compression as the Generative Mechanism of Thin Context

The previous two sections have established what thin context is and what domain-specific epistemic constraints are. This section establishes the mechanism by which one generates the other. Thin context is not produced by random failure, by inadequate data, or by an engineering deficiency that better training procedures might correct. It is produced by the systematic operation of a specific process — one that is constitutive of how statistical language representation works, not incidental to it. The process is **semantic compression**: the mechanism by which domain-specific epistemic constraints are collapsed during linguistic representation, such that the resulting encoding carries the statistical properties required for distributional generalization but not the validation-mechanism-specific information required for authoritative inference within an epistemic domain.

Semantic compression is the epistemic dimension of the general compression process described in §2. Shannon compression operates on information in the abstract; semantic compression operates on the specific class of information that constitutes meaning within epistemic domains. The distinction matters because the information lost to Shannon compression is information in general — bits that the channel cannot carry at the required rate. The information lost to semantic compression is a specific and non-random subset: the constraint structure that validation mechanisms require. That structure is not merely underrepresented in the training distribution; it is underrepresented in a systematic direction that correlates with domain specificity, institutional embeddedness, and low surface-frequency. A legal conclusion stated explicitly in a statute or contract appears in far fewer documents than the same proposition stated informally. A clinical protocol's specific application to a patient presentation involves far fewer direct lexical signals than a general health discussion. The constraint-dense, validation-mechanism-relevant content of epistemic domains is precisely the content that distributional compression removes preferentially.

The mechanism operates at each stage of the representational pipeline described in §2. At the level of the distributional hypothesis, meaning is identified with co-occurrence pattern — which means that meaning-constituting constraints that do not manifest as co-occurrence regularities are invisible to the representation from the outset. At the level of embedding, high-dimensional meaning is projected into a lower-dimensional space in which domain-specific distinctions compete with general semantic distinctions for representational capacity. The adjudicative sense of *consideration* in contract law competes with all other senses of *consideration* in the training corpus, and the geometric position of the resulting vector reflects the aggregate distribution, not the domain-specific constraint. At the level of attention, the context window assigns weight to tokens based on distributional co-occurrence relevance — which is not the same as validation-mechanism relevance. An attention head operating on a legal query attends to what statistically co-occurs with legal language in the training distribution; it does not attend to what the adjudicative framework requires for a determination to be authoritative.

Bender and Koller's analysis of the form-meaning distinction in large language models identifies the theoretical core of this mechanism.[26] Their argument is that training on form — the distributional patterns of word sequences — does not give access to meaning in the grounded sense: the relationship between linguistic expressions and the states of affairs, social institutions, and inferential frameworks they represent. Semantic compression is the process by which that access is foreclosed at representation time. When the domain-specific constraint structure that a validation mechanism operates on is compressed away, the resulting rep-

[26] Emily M. Bender and Alexander Koller, "Climbing towards NLU: On Meaning, Form, and Understanding in the Age of Data," in *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, 5185–5198 (Online: Association for Computational Linguistics, 2020), https://doi.org/10.18653/v1/2020.acl-main.463. Cited above, §4, fn. 23.

resentation has the surface properties of domain-relevant language without the epistemic substructure that would make domain-relevant inference possible. The text looks right. The inference does not constitute an answer by the domain's standards.

The operational consequences are visible in empirical studies of generation quality. Maynez and colleagues' analysis of faithfulness and factuality in abstractive summarization demonstrates that generation systems optimized for fluency systematically produce content that is both coherent and unfaithful to the source — hallucinating details, substituting statistically plausible claims for accurate ones, and maintaining a confident tone throughout.[27] The pattern is not random error; it is the expected output of a compression regime that has optimized for what reads well rather than what can be validated. Summarization is a tractable test case for a broader phenomenon: any generation task that requires output to be evaluated against a standard outside the training distribution's fluency metrics will exhibit this pattern. Legal, clinical, and engineering reasoning are precisely such tasks.

Hallucination — the confident generation of factually unsupported or factually false content — is the operational signature of thin context at the output level, and Ji and colleagues' comprehensive survey of hallucination taxonomies and causes across natural language generation tasks provides the empirical inventory.[28] The survey's taxonomy of intrinsic hallucination (contradicting the source) and extrinsic hallucination (generating unsupported content) maps directly onto the thin context condition: both arise when the generation process has no access to the validation mechanism that would distinguish supported from unsupported, valid from invalid, in-domain from out-of-domain. The model produces content with distributional authority; the validation mechanism is absent.

Scale sharpens this dynamic in a specific and important way. Larger models produce more fluent, more structurally coherent, and more contextually appropriate responses across a wider range of queries. This generalization improvement is real and well-documented.[29] It does not correspond to a recovery of the domain-specific constraint information that semantic compression removed, because that information was removed at the representational level, not the decoding level. Scale improves the quality of the distributional approximation; it does not change the relationship between distributional approximation and validation-mechanism authority. The result is that semantic compression at scale produces confident misalignment with greater fluency and over a broader domain range — which is precisely the failure mode that Paper One identifies and which the HGC³AE² framework is designed to address.[30]

The connection to distribution shift is structural. Quionero-Candela and colleagues' analysis of dataset shift in machine learning formalizes the condition in which a model's training distribution differs from its deployment distribution in ways that invalidate its learned mappings.[31] Domain crossing is a form of covariate shift: the input distribution at deployment differs from the training distribution in the features that matter for the target task. Thin context is the epistemic instantiation of this structural condition: the deployment context requires validation-mechanism constraints that were not represented in the training distribution, and

[27] Joshua Maynez, Shashi Narayan, Bernd Bohnet, and Ryan McDonald, "On Faithfulness and Factuality in Abstractive Summarization," in *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, 1906–1919 (Online: Association for Computational Linguistics, 2020), https://doi.org/10.18653/v1/2020.acl-main.173.

[28] Ziwei Ji, Nayeon Lee, Rita Frieske, Tiezheng Yu, Dan Su, Yan Xu, Etsuko Ishii, Ye Jin Bang, Andrea Madotto, and Pascale Fung, "Survey of Hallucination in Natural Language Generation," *ACM Computing Surveys* 55, no. 12, Article 248 (March 2023): 1–38, https://doi.org/10.1145/3571730.

[29] Tom B. Brown et al., "Language Models Are Few-Shot Learners," in *Advances in Neural Information Processing Systems 33* (NeurIPS 2020), 1877–1901, https://arxiv.org/abs/2005.14165. Cited above, §2, fn. 13.

[30] Justin H. Kuiper, CISSP, "Mitigating Confident Misalignment in Agentic Systems: The HGC³AE² Framework," Non Sequitur Publishing, 2026, v1.0-preprint, SHA 6e0b127f, https://nonsequitur.tech/white-papers/hgc3ae2/. Cited above, §1, fn. 1.

[31] Joaquin Quionero-Candela, Masashi Sugiyama, Anton Schwaighofer, and Neil Lawrence, eds., *Dataset Shift in Machine Learning* (Cambridge, MA: MIT Press, 2009).

the model's response to that deployment context reflects the training distribution's approximation rather than the deployment context's requirements. The risk-stratification algorithm in the clinical example was trained on a population distribution in which cost correlated with health need; deployed in a sub-population where that correlation diverged, it produced thin context because the validation constraints operative in the deployment context — the clinical protocol and individual patient circumstances — were absent from its representation.

Semantic compression is therefore not a bug to be patched. It is the mechanism by which statistical language representation produces the distributional generalizability that makes large language models useful. The same mechanism produces thin context as a structural output whenever the deployment context requires validation-mechanism constraints that the compression has removed. §§8 and 9 examine the architectural and human interventions available for managing this structural condition. §7 establishes why those interventions are epistemic necessities rather than governance preferences by connecting the thin context condition to the specific failure modes that Paper One identifies in agentic systems operating at scale.

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6. Human Inference Under Thin Context

The failure of statistical inference under thin context does not imply the success of human inference. Humans are not reliable reasoners, and the cognitive science literature is rich in documented failure modes — heuristic biases, overconfidence, anchoring, and the full taxonomy that Tversky and Kahneman's foundational work established.[32] The claim this paper makes is more limited and more specific: when human inference fails under thin context, the failure has structure that admits correction, escalation, and governed remediation. When statistical inference fails under thin context, the failure has no internal detection mechanism, no self-correcting structure, and no escalation path. The asymmetry is not between reliable humans and unreliable machines; it is between failures that are epistemically correctable and failures that are not. Understanding why requires understanding how human inference succeeds when it does.

Kahneman's dual-process account of human cognition distinguishes two broadly characterized modes of inference: a fast, associative, pattern-matching mode (System 1) that operates on distributional similarity and learned heuristics, and a slower, deliberative, rule-governed mode (System 2) that operates by applying structured frameworks to specific problems.[33] The relevant observation for the thin context problem is that System 1 and statistical language models share the same fundamental operation — distributional pattern completion — and fail under thin context for the same fundamental reason: they operate on surface regularities without access to the validation-mechanism frameworks that constitute domain-specific authoritative answers. System 2, however, is precisely the capacity to invoke those frameworks when the situation demands it. A lawyer who recognizes that a question has crossed into adjudicative territory does not try harder to generate a statistically plausible answer; she shifts cognitive modes and applies the deliberative apparatus of legal reasoning. That shift is an epistemic operation — the recognition that the relevant validation mechanism is adjudication, not pattern matching — and it is not available to a system that has no representation of validation mechanisms as such.

Human error under thin context follows the same structural logic. The heuristic failures that

[32] Amos Tversky and Daniel Kahneman, "Judgment under Uncertainty: Heuristics and Biases," *Science* 185, no. 4157 (1974): 1124–1131, https://doi.org/10.1126/science.185.4157.1124.
[33] Daniel Kahneman, *Thinking, Fast and Slow* (New York: Farrar, Straus and Giroux, 2011).

Tversky and Kahneman document — representativeness, availability, anchoring, framing effects — are System 1 failures: the application of distributional pattern-matching in contexts that require deliberative rule application.[34] These failures are not random; they are systematic, predictable, and well-characterized. They are also correctable, because the validation mechanisms of the relevant domains provide external standards against which the error can be identified and the inference revised. A clinician who has relied on anchoring in an initial diagnosis can revise that diagnosis when clinical protocol is applied systematically to the case. A lawyer who has relied on analogical pattern-matching can be corrected when the governing statute is examined directly. The correction is possible because the validation mechanism exists independently of the inference that failed, and the human can be directed to apply it. No equivalent correction path is available for a language model whose output reflects distributional compression rather than domain validation.

The human factors literature documents this asymmetry at the level of human-automation interaction. Trust calibration research shows that human reliance on automated systems is governed by domain-specific expectations of system competence — expectations that, when miscalibrated, produce the most dangerous failure modes precisely when system confidence and system reliability diverge.[35] This is a behavioral description of what thin context produces at the operational level: a system that asserts with high confidence in conditions where its outputs cannot be validated. Parasuraman and Riley's analysis of automation-induced complacency identifies the failure mode that results when users extend trust past the competence boundary of the automated system — a boundary that, under thin context, the system itself cannot signal.[36] The human's ability to recognize that boundary, when it functions, is the capacity for epistemic domain awareness: the recognition that the query has crossed into territory where the system's validation mechanism is inadequate and human deliberative judgment is required.

Context-switching and domain-boundary recognition are not uniformly reliable human capacities. They depend on domain expertise, training, and the institutional structures that make validation mechanisms visible. A lawyer who has been trained to recognize when a fact pattern implicates force majeure doctrine, and to distinguish that recognition from mere distributional similarity to contract language, is exercising an epistemic skill that took years of disciplined practice to develop. The same lawyer, asked a clinical question at the boundary of her expertise, may not recognize the domain boundary and may default to System 1 pattern matching. This is not a failure of the general capacity but a failure of its application to unfamiliar validation mechanisms — and it underscores that human epistemic authority is not a generic resource but a domain-specific one, dependent on the specific structures of training and institutional practice that make each domain's validation mechanism legible.

The *force majeure* example reaches its clearest resolution here. The lawyer's advantage over the language model in answering the question "does this event qualify as *force majeure* under this clause?" is not that she has access to more legal text. It is that she has access to the adjudicative framework as a structured procedure — she knows what questions to ask (what does governing law require? what is the factual record? how has this clause language been interpreted in analogous cases?), what answers would constitute progress toward an authoritative determination, and when to seek judgment from an authority empowered to make that determination binding. The language model has distributional patterns; the lawyer has the procedure. The physician evaluating the risk-stratification algorithm's output has, similarly, the clinical protocol, the specific patient relationship, and the institutional standing to make a clinical determination that overrides the algorithm's statistical output. Both have, in the

[34] Tversky and Kahneman, "Judgment under Uncertainty," cited above, fn. 32.

[35] Kevin Anthony Hoff and Masooda Bashir, "Trust in Automation: Integrating Empirical Evidence on Factors That Influence Trust," *Human Factors* 57, no. 3 (2015): 407–434, https://doi.org/10.1177/0018720814547570. Cited above, §1, fn. 4.

[36] Raja Parasuraman and Victor Riley, "Humans and Automation: Use, Misuse, Disuse, Abuse," *Human Factors* 39, no. 2 (1997): 230–253, https://doi.org/10.1518/001872097778543886.

terms of this paper, access to the validation mechanism that the compressed representation lacks. §8 examines what architectural interventions can do to close that gap, and §9 argues that closing it fully — for the class of cases where validation mechanisms are irreducibly institutional — requires human authority that no architecture replaces.

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7. From Thin Context to Confident Misalignment: The Bridge to HGC³AE²

The preceding five sections have developed two bodies of analysis that now converge. Sections 2 through 5 established the mechanism: language representation is systematic lossy compression; domain-specific epistemic constraints are what the compression removes preferentially; thin context is the condition that results; semantic compression is the process that generates it. Sections 3 and 6 established the stakes: epistemic domains constitute truth through validation mechanisms that the compression strips, and human inference succeeds where statistical inference fails because humans can invoke those mechanisms, imperfectly but correctably. The bridge this section draws is causal and structural, not analogical: thin context is not a condition that resembles the failure modes Paper One identifies. It is the epistemic precondition from which those failure modes emerge with structural necessity.

Paper One's taxonomy of failure modes in agentic systems — confident misalignment, context integrity failure, communication asymmetry, silent degradation, and execution–design divergence — reads differently once thin context is in view.[37] Confident misalignment is the condition in which the system produces output that is coherent, plausible, and presented with apparent authority, yet wrong relative to verified state or user intent. In the terms of this paper, confident misalignment is what thin context looks like at the output level: the system has operated on a representation from which domain-specific validation constraints have been removed, generated a distributional approximation in their place, and produced that approximation with the fluency and confidence that scale provides. The misalignment is confident because the compression that generated it also removed the signal that would indicate its own inadequacy. A system that lacks the domain constraints necessary to know that it does not know cannot hesitate.

Context integrity failure — the condition in which the assembled context does not accurately represent authorized state or real intent — is thin context entering the agentic system's operational loop. Agentic systems depend on assembled context: prompt history, retrieved documents, tool outputs, memory stores, and inferred task state. If that assembled context fails to supply the domain-specific epistemic constraints relevant to the system's next action, the system operates in thin context for that action, generating a distributional approximation of what the validation mechanism would require rather than satisfying the mechanism's actual requirements. The context is not corrupt; it is epistemically thin. And because the system cannot detect the thinness — cannot, without architectural intervention, compare its representation against an external domain validation standard — the integrity failure is silent.

Communication asymmetry is the user-side manifestation of the same condition. The human supplies intent, domain expertise, and contextual authority; the system infers distributional analogs. When the user's intent implicates a domain-specific validation mechanism — when the user is asking a legal question, a clinical question, an engineering question, or a governance question — the system's distributional inference is operating in thin context with re-

[37] Justin H. Kuiper, CISSP, "Mitigating Confident Misalignment in Agentic Systems: The HGC³AE² Framework," Non Sequitur Publishing, 2026, v1.0-preprint, SHA 6e0b127f, https://nonsequitur.tech/white-papers/hgc3ae2/. Cited above, §1, fn. 1. The taxonomy of failure modes appears in §2 of Paper One.

spect to exactly the constraint information the user's intent requires. The gap is not between what the user said and what the user meant, in the ordinary sense of miscommunication. It is between the validation-mechanism constraints that constitute the user's domain-specific intent and the distributional approximation that the system infers from the surface form of the user's expression. The user assumes the system has understood; the system has produced a statistically plausible continuation.

Silent degradation — the failure mode in which the system continues to operate with decreasing reliability without signaling the change — is precisely what thin context predicts when domain constraints are progressively stripped from an evolving context window. As the context accumulates distributional approximations and the validation-mechanism constraints recede, the system's outputs become progressively less epistemically grounded without any change in their surface fluency or apparent confidence. The degradation is silent because the signal that would indicate it — the presence or absence of domain validation constraints in the representation — is not a quantity the system monitors or surfaces. Execution–design divergence is the structural corollary: the system was designed under assumptions about the constraints that would be available at runtime, and those assumptions are violated by the thin context that the compression regime produces at deployment.

The agentic system literature provides its own independent corroboration of the thin context account. Amodei and colleagues' taxonomy of AI safety failure modes includes reward hacking and negative side effects — conditions in which the system optimizes for a proxy that correlates with the intended objective in the training distribution but diverges from it at deployment.[38] This is precisely the semantic compression dynamic: the proxy is a distributional approximation of the validation-mechanism requirement, adequate in the aggregate but failing at the domain boundary where the approximation is no longer sufficient. Park and colleagues' generative agents — autonomous agents operating on assembled context — exhibit behavior that degrades when the context they assemble fails to ground their inferences in the specific circumstances of their deployment environment.[39] They are operating in epistemically thin conditions whenever their assembled context has stripped the constraint structure relevant to their next action.

The implication for HGC^3AE^2 is direct and the direction of implication is important. The framework is not a response to a behavioral observation — "agentic systems fail, so let us add governance." It is a response to a structural analysis: thin context is the epistemic substrate from which agentic failure modes are generated, and addressing thin context requires the specific components the framework supplies. The Curated Context component (C^3) addresses thin context by reintroducing domain-specific constraints into the representation at inference time — replacing distributional approximation with grounded, validated, domain-relevant content. The Human authority component (H) addresses thin context by maintaining epistemic authority at the domain boundaries where validation mechanisms cannot be computationally approximated. The Governance component (G) addresses thin context by enforcing the institutional structures — permissions, escalation paths, accountability mechanisms — that keep domain validation requirements in force across the operational lifecycle. The framework is not a governance preference. It is what the epistemic analysis requires.

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[38] Dario Amodei, Chris Olah, Jacob Steinhardt, Paul Christiano, John Schulman, and Dan Mané, "Concrete Problems in AI Safety," arXiv:1606.06565, 2016, https://arxiv.org/abs/1606.06565.

[39] Joon Sung Park, Joseph O'Brien, Carrie Cai, Meredith Morris, Percy Liang, and Michael Bernstein, "Generative Agents: Interactive Simulacra of Human Behavior," arXiv:2304.03442, 2023, https://arxiv.org/abs/2304.03442.

8. Architectural Interventions

If thin context is generated structurally by semantic compression, then addressing it architecturally requires reversing — at least partially and at least at inference time — the compression process that produced it. The domain-specific epistemic constraints that the compression removed must be reintroduced into the representation before inference proceeds. This is the architectural principle: not better compression, not larger models, not more training data applied to the same distributional objective, but the reintroduction of the constraint structure that the compression stripped. Every architectural intervention that reduces thin context achieves what it achieves by approximating this principle. Every intervention that fails does so by failing to reintroduce the relevant constraint class.

Retrieval-augmented generation represents the most direct architectural response to thin context currently in widespread deployment.[40] The core mechanism is straightforward: rather than relying solely on the compressed representation that the language model has internalized, the system retrieves domain-relevant content at inference time and incorporates it into the context window alongside the query. This reintroduces grounded domain content that the pre-training compression had either not seen or had compressed to distributional statistical form. When the retrieved content carries domain-specific epistemic constraints — the governing statute for a legal query, the applicable clinical guideline for a medical query, the relevant engineering specification for a technical query — it reduces thin context by supplying what the validation mechanism requires, at least in part. The crucial qualifier is "at least in part": retrieval reduces thin context in proportion to the epistemic relevance of what is retrieved, not merely in proportion to the volume of domain text retrieved. Retrieving more contract language does not reduce thin context for a force majeure determination if the specific governing law and applicable precedent are not among the retrieved content.

Chain-of-thought prompting approaches thin context reduction from a different angle.[41] By eliciting explicit intermediate reasoning steps from the model, structured prompting makes the inference path visible and creates decision points at which domain-specific constraint checking can be inserted. The technique does not reintroduce domain constraints directly; it creates a representational surface where those constraints can be applied by human oversight or further architectural validation. To the extent that the model's intermediate steps surface assumptions that can be checked against domain validation requirements — does this intermediate legal conclusion follow from the governing statute? does this intermediate clinical inference accord with the applicable protocol? — chain-of-thought reduces effective thin context by creating the structure that makes human epistemic authority applicable at the right moments rather than only at the final output.

Domain-constrained generation — including vocabulary restriction, retrieval filtering, domain-specific fine-tuning, and constrained decoding — reduces thin context by limiting the distributional space within which generation operates. If the generation is constrained to vocabulary and constructions that are licensed by a specific domain's validation framework, the probability of producing outputs that violate that framework's constraints decreases. Fine-tuning on domain-specific data can, under favorable conditions, shift the model's distributional priors toward the constraint structure relevant to the target domain. These interventions are partial and do not eliminate thin context for the class of cases where validation-mechanism authority is constitutively institutional — where being valid is not a property of the text but a property of the process by which the text has been evaluated by an

[40] Patrick Lewis, Ethan Perez, Aleksandra Piktus, Fabio Petroni, Vladimir Karpukhin, Naman Goyal, Heinrich Küttler, Mike Lewis, Wen-tau Yih, Tim Rocktäschel, Sebastian Riedel, and Douwe Kiela, "Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks," in *Advances in Neural Information Processing Systems 33* (NeurIPS 2020), 9459–9474, https://arxiv.org/abs/2005.11401.

[41] Jason Wei, Xuezhi Wang, Dale Schuurmans, Maarten Bosma, Brian Ichter, Fei Xia, Ed Chi, Quoc V. Le, and Denny Zhou, "Chain-of-Thought Prompting Elicits Reasoning in Large Language Models," in *Advances in Neural Information Processing Systems 35* (NeurIPS 2022), https://arxiv.org/abs/2201.11903.

authorized entity. A well-constrained domain fine-tuned model can produce text that looks like a valid legal conclusion; it cannot produce a conclusion that is legally valid, because legal validity is not a textual property.

Explainability infrastructure provides a different form of thin context management: not elimination but detection.[42] When a model's inference path is made visible through attribution methods, saliency maps, or natural language explanations of the reasoning chain, human overseers can identify the moments at which domain-specific constraint application is missing — the points in the reasoning where the model has made a distributional jump that the relevant validation mechanism would not license. Explainability does not itself reintroduce the missing constraints; it makes the absence of those constraints visible at a level of granularity that allows human authority to be applied at the right moment. In this sense, explainability infrastructure is a thin context diagnostic tool, and its value is greatest in high-stakes deployment contexts where human authority is the backstop and early detection of constraint absence is operationally critical.

The cumulative picture these interventions produce is one of partial reduction rather than elimination. Each class of intervention addresses one aspect of the thin context condition: retrieval reintroduces grounded domain content; chain-of-thought creates constraint-checkable intermediate steps; domain constraint reduces the distributional space; explainability makes constraint absence detectable. None of them reconstructs the full validation mechanism of the target domain, because that reconstruction would require encoding the institutional authority, procedural requirements, and domain-specific expertise that the mechanism depends on — and those are not properties of any text representation. The clinical deployment of retrieval augmentation illustrates the residual limit directly: a system that retrieves relevant clinical guidelines reduces thin context for the class of queries that those guidelines address, but cannot reduce thin context for the physician's judgment about whether the retrieved guidelines are the operative ones for this specific patient at this specific moment in treatment.[43] That judgment requires clinical authority — the domain's validation mechanism — which the architecture can support but not replace.

This is the structural argument for why architectural interventions are necessary but not sufficient conditions for addressing thin context in high-stakes domains. They reduce the frequency and severity of thin context conditions; they do not eliminate the structural dependence on human epistemic authority at the domain boundaries where validation mechanisms are irreducibly institutional. §9 examines what the human side of this equation requires.

9. Human Interventions and Epistemic Authority

The architectural interventions examined in §8 reduce thin context; they do not eliminate it. At the boundary where validation mechanisms are constitutively institutional — where the validity of a determination depends not on the properties of any text but on the authority and procedure of the entity making it — no retrieval system, no constrained decoding scheme, and no explainability mechanism supplies what is missing. What is missing is not retrievable content or a better decoding objective. It is the epistemic authority to constitute a valid

[42] Alejandro Barredo Arrieta, Natalia Díaz-Rodríguez, Javier Del Ser, Adrien Bennetot, Siham Tabik, Alberto Barbado, Salvador Garcia, et al., "Explainable Artificial Intelligence (XAI): Concepts, Taxonomies, Opportunities and Challenges Toward Responsible AI," *Information Fusion* 58 (2020): 82–115, https://doi.org/10.1016/j.inffus.2019.12.012.

[43] Eric J. Topol, "High-performance medicine: the convergence of human and artificial intelligence," *Nature Medicine* 25 (2019): 44–56, https://doi.org/10.1038/s41591-018-0300-7. Cited above, §3, fn. 20.

determination within the domain. The human side of the equation, as §8 promised to examine, is the supply side for that authority.

The literature on human-centered AI has articulated the case for human involvement in AI systems primarily as a design preference: systems should be oriented toward human values, human oversight should be maintained as a matter of governance prudence, and human agency should be preserved against inappropriate delegation.[44] These are valid claims, and the design principles they underwrite — keep the human in the loop, maintain meaningful oversight, ensure interpretability — are widely reflected in institutional AI governance frameworks. What this paper argues is that they are underspecified. Keeping a human in the loop is not sufficient if the human in the loop lacks the domain-specific epistemic authority that the thin-context condition requires. A non-expert human in the review position of a clinical AI system does not supply the physician's validation authority; the ergonomic form of human-in-the-loop is present without its epistemic content.

Russell's account of the AI control problem frames value alignment as the central challenge: AI systems must be designed to pursue objectives that accurately reflect human values, and the difficulty lies in specifying and maintaining that alignment across the range of contexts in which the system will operate.[45] Paper Two's analysis reframes this challenge in epistemic terms. Values, in the relevant class of high-stakes domains, are not free-floating preferences to be specified and internalized; they are epistemic constraints constituted by the domain's validation mechanisms. Legal value — the authority of a binding determination — is constituted by adjudication, not by training. Clinical value — the validity of a treatment decision — is constituted by clinical protocol applied by an authorized practitioner, not by distributional preference. The alignment problem, in this framing, is a thin-context problem: AI systems operating under semantic compression lack access to the constraint structure in which domain-specific values are constituted, and recovering that alignment requires not better objective specification but access to the validation mechanisms themselves.

Ryan's critique of human-centered AI frameworks identifies a genuine limitation that the epistemic account must acknowledge: human judgment is not reliably better than automated inference, and frameworks that invoke human authority without specifying the conditions under which that authority is epistemically warranted risk substituting one failure mode for another.[46] The thin-context analysis provides the specificity that Ryan's critique demands. The claim is not that human judgment is superior in general; it is that human failure under thin context has structure that admits correction, while statistical inference failure under thin context does not. The physician who makes a clinical error under time pressure can have that error identified and corrected by the clinical protocol applied at a subsequent decision point. The protocol — the validation mechanism — exists independently of the individual inference and provides the external standard against which error is detectable and correctable. No equivalent correction path exists for a distributional system that has produced a confident output from a representation from which the relevant protocol was absent. This structural asymmetry, not a general claim about human reliability, is the epistemic ground for human authority at domain boundaries.

The governance dimension of human epistemic authority is what the G component of HGC^3AE^2 supplies. Governance in the thin-context framework is not primarily a set of compliance requirements or risk mitigation procedures; it is the institutional architecture that maintains epistemic authority at domain boundaries across the operational lifecycle. Batool and colleagues' systematic literature review of AI governance frameworks documents the convergence across institutional, regulatory, and industry governance initiatives on human over-

[44] Ben Shneiderman, *Human-Centered AI* (Oxford: Oxford University Press, 2022).

[45] Stuart Russell, *Human Compatible: Artificial Intelligence and the Problem of Control* (New York: Viking, 2019).

[46] Mark Ryan, "We're Only Human After All: A Critique of Human-Centred AI," *AI & Society* 40 (2025): 1303–1319, https://doi.org/10.1007/s00146-024-01976-2.

sight requirements as a core design principle.[47] The DoD's articulation of AI ethics principles identifies human judgment as the non-negotiable requirement at the boundary between automated recommendation and binding action.[48] The EU AI Act's high-risk category regime operationalizes this requirement as a regulatory mandate: human oversight must be maintained in AI applications where errors could cause significant harm, specifically because those are the applications where thin-context failure has the highest stakes.[49] The OECD AI Recommendation's human-centric principles establish the same requirement at the international policy level.[50] Floridi and colleagues' AI4People framework provides the ethical grounding: the allocation of epistemic authority between automated and human agents should be governed by the principle of epistemic subsidiarity — decisions should be made at the level of the agent whose validation mechanism is adequate for the domain of the decision.[51]

The NIST AI Risk Management Framework operationalizes these governance requirements at the organizational level, and Paper Two's analysis provides the epistemic foundation for why those requirements are structurally necessary rather than merely precautionary.[52] The framework's guidance on human review of AI outputs — particularly in high-stakes decision contexts — is not an ergonomic add-on; it is the institutional mechanism by which epistemic authority is returned to the representation at the moment the compressed model output is evaluated against the domain's validation standard. Prunkl's analysis of human autonomy under AI delegation identifies the failure mode that the thin-context account predicts: when AI systems operate under thin context and that thin-context output is the input to human decision-making, delegated autonomy is autonomy delegated to a system that lacks the epistemic constraints necessary for valid inference.[53] The delegation does not eliminate thin context; it relocates it into the human-AI handoff, where it may be even less visible. Taylor and colleagues' account of human-centric AI design identifies community-level epistemic authority — the domain knowledge distributed across practitioners, institutions, and social processes — as the resource that AI systems must be designed to elicit rather than replace.[54]

The institutional design question is what Schuett and colleagues address: what form must epistemic authority structures take to function as effective epistemic governance mechanisms rather than nominal oversight processes?[55] Their analysis of AI ethics board design maps onto the thin-context condition directly: effective oversight structures must include domain-specific expertise — the bearers of the relevant validation mechanisms — and must be positioned at the decision points where thin-context outputs are most likely to be produced. A governance structure that operates only at the level of general AI principles, without domain-specific validators at the domain boundary, provides the form of epistemic authority without its function. §10 examines the implications of this analysis for how the field evaluates AI

[47] Amna Batool, Didar Zowghi, and Muneera Bano, "AI Governance: A Systematic Literature Review," *AI and Ethics* 5 (2025): 3265–3279, https://doi.org/10.1007/s43681-024-00653-w.

[48] U.S. Department of Defense, "DoD Adopts Ethical Principles for Artificial Intelligence," February 24, 2020, https://www.defense.gov/News/Releases/Release/Article/2091996/.

[49] European Parliament and Council of the European Union, Regulation (EU) 2024/1689 (Artificial Intelligence Act), *Official Journal of the European Union* L, 2024/1689 (July 12, 2024), https://eur-lex.europa.eu/eli/reg/2024/1689/oj.

[50] Organisation for Economic Co-operation and Development, *Recommendation of the Council on Artificial Intelligence*, OECD/LEGAL/0449, adopted May 22, 2019; amended May 3, 2024, https://legalinstruments.oecd.org/en/instruments/OECD-LEGAL-0449.

[51] Luciano Floridi, Josh Cowls, Monica Beltrametti, Raja Chatila, Patrice Chazerand, Virginia Dignum, Christoph Luetge, Robert Madelin, Ugo Pagallo, Francesca Rossi, Burkhard Schafer, Peggy Valcke, and Effy Vayena, "An Ethical Framework for a Good AI Society: Opportunities, Risks, Principles, and Recommendations," *Minds and Machines* 28, no. 4 (2018): 689–707, https://doi.org/10.1007/s11023-018-9482-5.

[52] National Institute of Standards and Technology, *Artificial Intelligence Risk Management Framework (AI RMF 1.0)*, NIST AI 100-1 (Gaithersburg, MD, 2023), https://doi.org/10.6028/NIST.AI.100-1. Cited above, §1, fn. 2.

[53] Carina Prunkl, "Human Autonomy at Risk? An Analysis of the Challenges from AI," *Minds and Machines* 34 (2024), https://doi.org/10.1007/s11023-024-09665-1.

[54] Randon R. Taylor, Bessie O'Dell, and John W. Murphy, "Human-Centric AI: Philosophical and Community-Centric Considerations," *AI & Society* 39 (2024): 2417–2424, https://doi.org/10.1007/s00146-023-01694-1.

[55] Jonas Schuett, Anka Reuel, and Alexis Carlier, "How to Design an AI Ethics Board," *AI and Ethics* 5, no. 2 (2024): 863–881, https://doi.org/10.1007/s43681-023-00409-y.

systems: if the relevant failure mode is thin context at domain boundaries, and if the standard evaluation regime does not assess behavior at those boundaries, the standard regime is measuring the wrong thing.

10. Implications for Evaluation

The standard evaluation regime for NLP systems measures the wrong thing. Benchmark suites such as GLUE, SuperGLUE, and their successors assess average-case distributional generalization: how well does the model produce statistically appropriate outputs across a broad range of surface-form queries drawn from the training distribution? This is a technically well-defined and practically useful measurement, and the improvements it has tracked over the past decade are real. It is not a measurement of constraint preservation across domain boundaries. A model that achieves state-of-the-art performance on every available benchmark may be producing thin context whenever it operates on legal, clinical, or engineering queries — and the benchmarks will not register the failure. The evaluation framework is calibrated to the distributional objective, not to the validation-mechanism requirements that the deployment context imposes.

Ribeiro and colleagues' CheckList framework represents the most principled available attempt to extend behavioral evaluation beyond accuracy on held-out distributional data.[56] The framework introduces capability-specific test types — minimum functionality tests, invariance tests, directional expectation tests — designed to probe whether a model's output behavior reflects coherent underlying capacities rather than surface-form pattern matching. This is a genuine advance: it tests what the model does rather than merely how often its output matches a distributional target. What it does not test, and cannot test from within the behavioral probing paradigm, is whether the model's output satisfies the validation requirements of an external domain authority. A model that passes all invariance tests for legal queries has demonstrated consistent behavior; it has not demonstrated behavior that a court would find legally valid. CheckList identifies the gap between average-case accuracy and capability-specific behavior; it does not close the gap between capability-specific behavior and domain-specific epistemic validity.

Ji and colleagues' hallucination survey provides the most comprehensive empirical inventory of the condition that thin context produces at the output level.[57] The taxonomy of intrinsic and extrinsic hallucination documents the frequency, domain distribution, and surface characteristics of outputs that are unfaithful or unsupported. From the perspective of this paper, the survey describes the symptom; the thin-context condition is the structural explanation. Evaluation frameworks that target hallucination reduction without targeting thin context at domain boundaries are addressing the symptom at the distributional level while leaving the structural cause in place. The Obermeyer et al. finding provides the operational illustration: a risk-stratification algorithm with high average-case accuracy produced systematically biased outputs at the domain boundary where the correlation between its proxy variable and its validation target diverged from the training distribution.[58] Average-case evaluation did not

[56] Marco Tulio Ribeiro, Tongshuang Wu, Carlos Guestrin, and Sameer Singh, "Beyond Accuracy: Behavioral Testing of NLP Models with CheckList," in *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, 4902–4912 (Online: Association for Computational Linguistics, 2020), https://aclanthology.org/2020.acl-main.442.

[57] Ziwei Ji, Nayeon Lee, Rita Frieske, Tiezheng Yu, Dan Su, Yan Xu, Etsuko Ishii, Ye Jin Bang, Andrea Madotto, and Pascale Fung, "Survey of Hallucination in Natural Language Generation," *ACM Computing Surveys* 55, no. 12, Article 248 (March 2023): 1–38, https://doi.org/10.1145/3571730. Cited above, §5, fn. 28.

[58] Ziad Obermeyer, Brian Powers, Christine Vogeli, and Sendhil Mullainathan, "Dissecting Racial Bias in an Algorithm Used to Manage the Health of Populations," *Science* 366, no. 6464 (October 25, 2019): 447–453,

detect the failure because average-case performance was genuinely high. Domain-boundary performance was what mattered clinically, and it was not measured.

Explainability infrastructure, as examined in §8, offers an evaluation resource that the standard benchmark paradigm does not exploit. Arrieta and colleagues' taxonomy of explainability methods identifies attribution, saliency, and natural language explanation techniques that make the model's inference path partially visible.[59] When evaluation incorporates explainability analysis — examining not just whether the output is correct but whether the inference path traverses the constraint structure that the validation mechanism requires — it can identify thin-context conditions directly rather than inferring them from output error rates. A legal output that is produced without engaging the governing statute's structure, even if the output happens to be correct, is evidence of thin context that an explainability-augmented evaluation regime can surface. The NIST Generative AI Profile operationalizes this approach at the risk-management level, identifying domain-specific risk assessment for generative AI applications as a governance requirement rather than an optional analysis.[60]

What epistemically honest evaluation of NLP systems in high-stakes domains requires is a four-part extension of the current paradigm. First, evaluation must test explicitly at domain boundaries — not just on general-knowledge queries but on legal, clinical, and engineering queries where validation mechanisms are operative, and specifically on the queries where the statistical proxy diverges from the validation-mechanism requirement. Second, evaluation must assess constraint preservation rather than merely distributional accuracy: does the output engage the governing law, the applicable protocol, the relevant specification, or does it produce a distributional approximation that lacks that engagement? Third, evaluation must test human-in-the-loop behavior under thin context: when the system produces thin-context output, does the system architecture correctly escalate to human epistemic authority, or does it produce confident misalignment that the oversight structure will not detect? Fourth, evaluation must assess calibration under domain-crossing conditions — whether the system's expressed confidence tracks its actual constraint availability rather than its distributional fluency.

Bender and colleagues' analysis of the risks of large-scale language models identifies the core evaluative blind spot directly: because distributional training and distributional evaluation share the same metric — surface-form statistical coherence — systems optimized on distributional objectives can achieve high evaluation scores while systematically failing at tasks that require grounded constraint engagement.[61] The evaluation framework is not independent of the training framework; both are calibrated to the same objective. Reforming evaluation requires exiting that shared metric and assessing behavior against an external standard — which, in high-stakes domains, means assessing behavior against the validation requirements of the domain itself. Prunkl's analysis of human autonomy under AI delegation identifies what is at stake operationally: when mismeasured systems are trusted with high-stakes determinations, the human autonomy and epistemic authority that the governance structure is designed to preserve are delegated to systems whose failures are invisible to the evaluation regime that certified them.[62] §11 draws the paper's conclusion: the epistemic architecture of language is the source of this condition, and addressing it is not a matter of better engineering within

https://doi.org/10.1126/science.aax2342. Cited above, §3, fn. 19.

[59] Alejandro Barredo Arrieta, Natalia Díaz-Rodríguez, Javier Del Ser, Adrien Bennetot, Siham Tabik, Alberto Barbado, Salvador Garcia, et al., "Explainable Artificial Intelligence (XAI): Concepts, Taxonomies, Opportunities and Challenges Toward Responsible AI," *Information Fusion* 58 (2020): 82–115, https://doi.org/10.1016/j.inffus.2019.12.012. Cited above, §8, fn. 42.

[60] National Institute of Standards and Technology, *Artificial Intelligence Risk Management Framework: Generative Artificial Intelligence Profile*, NIST AI 600-1 (Gaithersburg, MD, 2024), https://doi.org/10.6028/NIST.AI.600-1.

[61] Emily M. Bender, Timnit Gebru, Angelina McMillan-Major, and Shmargaret Shmitchell, "On the Dangers of Stochastic Parrots: Can Language Models Be Too Big?" in *Proceedings of the 2021 ACM Conference on Fairness, Accountability, and Transparency*, 610–623 (FAccT 2021), https://doi.org/10.1145/3442188.3445922.

[62] Carina Prunkl, "Human Autonomy at Risk? An Analysis of the Challenges from AI," *Minds and Machines* 34 (2024), https://doi.org/10.1007/s11023-024-09665-1. Cited above, §9, fn. 53.

the existing paradigm but of understanding what the paradigm structurally cannot do.

11. Conclusion

The limitation this paper has analyzed is not a limitation of intelligence. It is not a limitation of compute, of scale, or of the alignment objective in its narrow technical sense. It is a limitation of the epistemic architecture of language itself — the structure through which linguistic representation encodes meaning by distributional compression, and through which that compression systematically strips the domain-specific constraint structure that epistemic domains require for authoritative inference. This is not a contingent engineering deficiency that better training procedures or larger models will correct. It is the structural output of a representational process whose generalizability and whose domain-specific constraint loss are produced by the same mechanism. The condition that results — thin context — is what the analysis has named, defined, and traced through its consequences.

The analytical chain this paper has constructed runs from first principles to operational architecture. Shannon's noisy channel theorem establishes that high-noise, limited-capacity transmission requires lossy compression; the distributional hypothesis operationalizes that compression for linguistic meaning; the resulting representations achieve generalizability by encoding statistical regularities rather than validation-mechanism constraints. Epistemic domains — law, medicine, engineering, science — constitute truth not through statistical regularities but through specific validation procedures: adjudication, clinical protocol, engineering specification, empirical replication. Semantic compression strips those procedures from the representation. The result is thin context: the condition in which a system can produce fluent, confident, structurally complete outputs for domain-specific queries while lacking the constraint structure that the domain's validation mechanism requires to constitute an authoritative answer. Confident misalignment is the operational signature of thin context at the output level; context integrity failure, communication asymmetry, silent degradation, and execution–design divergence are its agentic system analogs.[63] HGC^3AE^2 is not a governance preference layered atop a capable system. It is what the epistemic analysis structurally requires.

The implications for the field follow directly. Evaluation reform is not an optional enhancement to the standard benchmark paradigm; it is a baseline requirement given what this paper has demonstrated. Benchmarks that measure average-case distributional generalization are calibrated to the same objective as the training process, and they will not detect thin-context failure at domain boundaries unless they are redesigned to test precisely at those boundaries. Architectural discipline — the C^3 component's commitment to curated, validated, domain-relevant context reintroduction at inference time — is not an ergonomic improvement; it is the mechanism by which the stripped constraint structure is returned to the representation where the deployment context requires it. Epistemic authority preservation — the H and G components' maintenance of human validation authority at domain boundaries and institutional governance structures across the operational lifecycle — is not a precautionary supplement; it is the supply side of the validation mechanism that no architecture replaces.

Kuhn's account of scientific paradigms illuminates the broader condition.[64] Normal science proceeds within a paradigm that defines the relevant questions, the valid methods, and the

[63] Justin H. Kuiper, CISSP, "Mitigating Confident Misalignment in Agentic Systems: The HGC^3AE^2 Framework," Non Sequitur Publishing, 2026, v1.0-preprint, SHA 6e0b127f, https://nonsequitur.tech/white-papers/hgc3ae2/. Cited above, §1, fn. 1.

[64] Thomas S. Kuhn, *The Structure of Scientific Revolutions*, 2nd ed. (Chicago: University of Chicago Press, 1970). Cited above, §3, fn. 15.

standards of success. Thin context is not a normal-science problem: it is not soluble within the distributional paradigm because the distributional paradigm constitutes meaning in a way that structurally excludes the constraint information that epistemic domains require. Addressing it requires operating at the paradigm boundary — understanding what the distributional framework structurally cannot do and designing the architectural, institutional, and human components that supply what the framework omits. Taylor and colleagues' account of human-centric AI as the mobilization of distributed community epistemic authority names what those components must ultimately mobilize: not a generic human override, but the specific, domain-trained, institutionally embedded epistemic authority that each validation mechanism requires.[65]

Russell frames the AI control problem as the problem of specifying what humans want: AI systems must pursue objectives aligned with human values, and the difficulty is encoding those values in a form the system can pursue.[66] This paper's analysis suggests that the framing, while correct as far as it goes, understates the epistemic depth of the problem. Values in high-stakes domains are not preferences to be specified and encoded; they are epistemic constraints constituted through the validation procedures of those domains — constituted through adjudication, clinical protocol, engineering review, scientific replication. The problem of AI control in epistemic domains is not the problem of specifying what humans want. It is the problem of preserving the epistemic constraints through which domains of human practice constitute what is true, valid, and permissible — and of ensuring that the systems operating in those domains either carry those constraints or operate under the authority of agents who do.

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[65] Randon R. Taylor, Bessie O'Dell, and John W. Murphy, "Human-Centric AI: Philosophical and Community-Centric Considerations," *AI & Society* 39 (2024): 2417–2424, https://doi.org/10.1007/s00146-023-01694-1. Cited above, §9, fn. 54.

[66] Stuart Russell, *Human Compatible: Artificial Intelligence and the Problem of Control* (New York: Viking, 2019). Cited above, §9, fn. 45.

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Appendix — v1.1 Errata Addendum

Added 2026-05-17 per Round-2 EDITORIAL-STAMP HOLD-substance verdict. The v1.0-preprint remains canonical at its original DOI; this v1.1 supplements with the corrections below.

Errata addendum for Paper Two v1.1

The following fenced block is the self-contained insertion for Paper Two. TM extracts to `intake/papers/<paper-two-slug>/v1.1-errata-addendum.md` at merge.

'markdown

addendum_for: "Paper Two — On Cognitive Nodes and HGC³AE²" addendum_version: "v1.1" addendum_date: "2026-05-17" trigger: "YKS-Spine-Binder#312 — epistemic-flaw case study" addendum_author: "Edna" —

§X. Epistemic Categories in HGC³AE² Memory — A Case Study from Operations

§X.1. The incident

On the evening of 2026-04-27, an agentic stack operating under the HGC³AE² framework produced a confident, structurally well-formed operational claim: that a set of papers (designated P4–P7 in the stack's internal tracking) were still in-flight and scheduled for a future sprint window. A human operator, running a live-verification probe against the publication surface (`nonsequitur.tech/pubs/`), discovered that P4–P7 had shipped two days earlier, on 2026-04-25. The claim was not approximately wrong. It was categorically stale.

The error did not originate in the LLM substrate. The model did not hallucinate a paper count or confuse document titles. It faithfully reported what its memory layer told it. The memory layer, in turn, faithfully stored a snapshot that had been accurate at write-time — and that no process had refreshed or expired since.

The operator's assessment was immediate and precise: **epistemic flaw**. Not a bug in the conventional sense. A category error in how the system's memory was structured, which the governing framework — HGC³AE² as written in v1.0 — had no mechanism to prevent.

This incident is worth documenting not because the failure was catastrophic (it was caught in seconds by a human probe) but because the failure was *architecturally silent*. Nothing in the system flagged the stale claim. The framework, as designed, approved the memory architecture that produced it. The gap is in the framework itself.

§X.2. What v1.0 said and what v1.0 missed

Paper Two v1.0 articulates the HGC³AE² framework as a governance discipline for human-directed agentic systems. The framework assigns specific responsibilities to each positional letter: Human governance (H), Governance structure (G), Curation (C), and so on through the remaining letters. `[TO BE VERIFIED AGAINST PAPER TWO v1.0]` The Curation layer (first C) is described as responsible for the quality and integrity of knowledge that enters the system's operating context. `[TO BE VERIFIED AGAINST PAPER TWO v1.0]`

What v1.0 does not do — anywhere — is distinguish between *kinds* of knowledge by their epistemic status. The framework speaks of curation as a quality gate but does not define what "quality" means when the same memory store contains both architectural decisions (true until explicitly refuted) and sprint-status snapshots (true only at the moment of capture). The Curation layer, as specified, would approve a memory file that intermixes both categories, because v1.0 provides no schema-level vocabulary to tell them apart. The framework was silent on the distinction. That silence is the gap.

The incident on 2026-04-27 did not reveal a framework violation. It revealed a framework *omission*. The system operated within HGC³AE² discipline and still produced a confidently wrong assertion — because the discipline, as written, lacked the epistemic category needed to prevent the mixing.

§X.3. The two-category discipline

We define two epistemic categories that any HGC³AE²-compliant memory layer must distinguish at the schema level:

Durable knowledge. Patterns, decisions, constraints, architectural commitments, learnings extracted from operational experience. Truth-status: *true-until-refuted*. A durable-knowledge entry remains load-bearing until an explicit governance act (a new ADR, an errata, a framework amendment) supersedes it. Durable knowledge decays — an architectural decision from six months ago may need re-examination — but the decay is slow, bounded, and governable through periodic review. The default posture toward a durable-knowledge entry at load time is: *trust, but verify on cadence.*

Ephemeral state. Current sprint, in-flight work items, open issue counts, today's publication surface, active persona assignments. Truth-status: *snapshot-at-write-time; stale-by-default thereafter*. An ephemeral-state entry is accurate only at the moment it was captured. Every subsequent load is a stale read unless the entry has been explicitly refreshed. The default posture toward an ephemeral-state entry at load time is: *distrust unless freshness is verified.*

The discipline requires three mechanical invariants:

1. **Schema separation.** Durable knowledge and ephemeral state MUST NOT coexist in a single undifferentiated memory store. Either they occupy separate files/tables, or each entry carries a mandatory `category: durable | ephemeral` tag that the loading mechanism respects.
2. **Load-time tagging.** When the agent's context window is populated from memory, every entry MUST surface its category. The agent must be able to distinguish, at reasoning time, between "this is an architectural constraint" and "this was the sprint status nine days ago."
3. **Freshness assertion for ephemeral entries.** Before an agent asserts any claim derived from an ephemeral-state entry, it MUST either (a) verify the entry's freshness against a live source, or (b) explicitly qualify the claim as potentially stale. Confident assertion of unverified ephemeral state is a framework violation under this amendment.

These invariants are necessary conditions. Individual implementations may impose additional discipline (e.g., automatic expiry of ephemeral entries after a configurable TTL, or mandatory refresh-on-load for entries older than one sprint cycle). The three invariants above are the floor.

§X.4. The framework amendment

We adopt a **clarifying clause** rather than a new letter or sub-letter. The HGC³AE² acronym is already dense; adding positional complexity for what is fundamentally a *refinement of an* *existing responsibility* would weaken the framework's mnemonic economy.

The amendment reads:

> **Clarifying clause — Epistemic-Category Discipline (appended to Curation).** The Curation layer (first C) is responsible not only for the quality of knowledge entering the system's operating context, but for the *epistemic categorization* of that knowledge. Specifically: Curation must enforce schema-level separation between durable knowledge (true-until-refuted) and ephemeral state (snapshot-at-write-time, stale-by-default). Memory architectures that intermix these categories without schema discrimination are non-compliant with HGC^3AE^2 Curation discipline, regardless of the quality of their individual entries.

This clause does not change what the first C *means*. It makes explicit what the first C *requires* — a requirement that v1.0's silence allowed compliant systems to overlook.

§X.5. Implications for downstream papers

This case study opens two forward threads.

First, the mechanical fix — schema-separated memory stores with category tagging and freshness discipline — is an architectural concern tracked at ADR-YKS-002 (Knowledge / State Separation) `[FORWARD — pending v1.0]`. ADR-YKS-002 is the canonical home for the implementation pattern. This errata addendum is the framework-level companion: it establishes *why* the separation is required under HGC^3AE^2, while the ADR establishes *how* a specific system implements it.

Second, Series 2 of the publication catalog (Managing Agentic Operations) should consider a dedicated paper on memory architecture for HGC^3AE^2-compliant systems — covering not only the durable/ephemeral split but broader questions of memory lifecycle, decay policy, and the relationship between agent memory and human-governed canon `[FORWARD — pending v1.0]`. The incident documented here is one failure mode; a systematic treatment would map the full surface.

§X.6. Operational receipt

The incident that motivated this addendum has a complete operational trail:

- **TM session date:** 2026-04-27 evening, during sprint-0427-0510.
- **Stale claim:** TM asserted P4–P7 were in-flight; P4–P7 had shipped 2026-04-25.
- **Live-verification trigger:** Human operator probed `nonsequitur.tech/pubs/`; mismatch surfaced immediately.
- **Operator assessment:** "epistemic flaw" — directed inclusion as Paper Two case study.
- **Architectural fix tracking:** ADR-YKS-002 (Knowledge / State Separation), yks-ops-hub#132.
- **Errata tracking:** YKS-Spine-Binder#312. '